Finance Dept ChatBot

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Problem Statement

- Finance major/minor requirements
- Course prerequisites
- Program pathways
- Scheduling and advising FAQs

Our Solution

A domain-specific **RAG (Retrieval Augmented Generation)** chatbot

- ✓ Rooted in official SCU documentation
- ✓ 24/7 instant academic advising
- ✓ Connects students to Drahmann Center when needed

Context of Data

Files from Jocelyn - Finance Dept

1. Finance Requirements List
2. Real Estate Minor Check Sheet
3. FAQ Advising Questions
4. Finance Prerequisites Flowchart
5. 2025-2026 Tentative Finance Course Offerings
6. Webpages for Finance Bot

Finance Requirement Checklist (Cohort 6_1_2025 and later) .XLSX

File Edit View Insert Format Data Tools Help

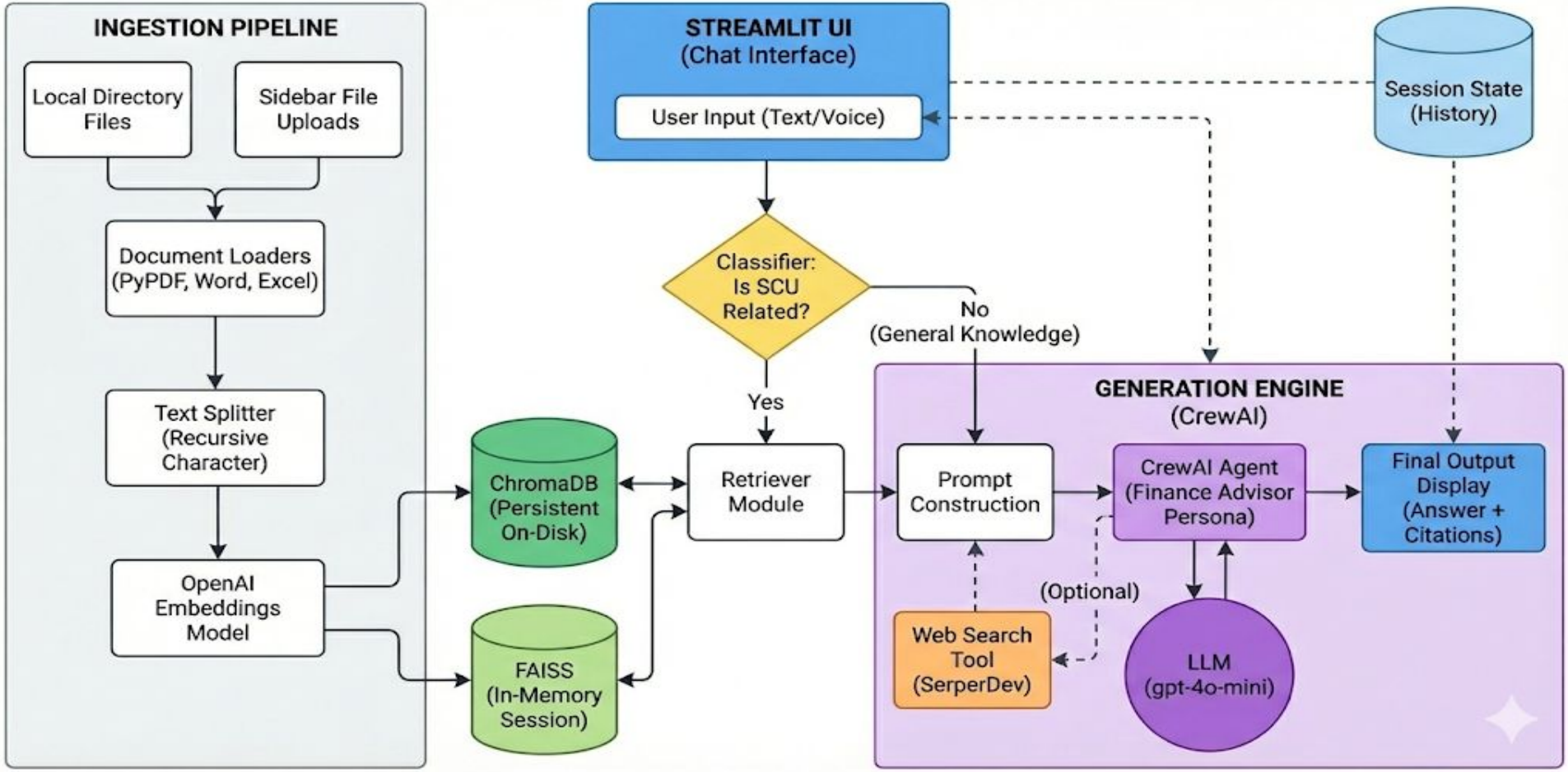
Menus 100% $ % .0 .00 123 Calibri 12

It is the responsibility of all students to know and fulfill the graduation requirements for their degree as outlined in the University

A	B	C	D
It is the responsibility of all students to know and fulfill the graduation requirements for their degree as outlined in the University Bulletin.			
		First Year Courses	**Recommended courses by year. Please note, this is a guide, as ea student's journey will vary**
Updated: 8/26/25		**Sophomore Year Courses**	
		Junior/Senior Year Courses	
		All Four Years	
University Core Requirements			
Foundations	**Status**	**Notes**	**Prerequisites**
Critical Thinking and Writing 1*		Required to Complete First Year	
Critical Thinking and Writing 2*		Required to Complete First Year	
Cultures and Ideas 1*		Required to Complete First Year	
Cultures and Ideas 2*		Required to Complete First Year	
2nd Language		Link to 2nd Language Info	
Math (MATH 11 or MATH 30)		**See BUSN Core**	
Religion, Thelogy, Cultures 1*			
Explorations	**Status**	**Notes**	**Prerequisites**
Ethics (MGMT 6 or PHIL 26)		**See BUSN Core**	
Civic Enagement (MGMT 162+ BUSN Ethics)		**See BUSN Core**	
Cultures and Ideas 3*		MGMT 80 only if taken at SCU	
Diversity			
Arts			
Natural Science and Lab			
Science, Technology, and Society (OMIS 34)		**See BUSN Core**	

Architecture

Model Architecture



Core Components

Configuration

API Key, set up KB Directory, index, chunk, build chroma and Faiss, Token, port number, Hosting (Ngrok)

Vectorization

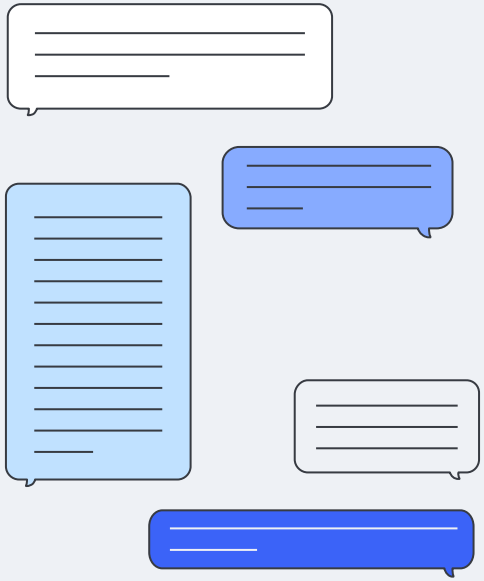
Encodes documents into vectors that power accurate retrieval during conversation

RAG

Retrieve context to generate relevant information

Front-end

Connects the backend logic to a simple, intuitive user interface



Helpers/Tools

Helper Functions

Load Doc to Directory

- Loading and Parsing using langchain (PyPDF,Word,Excel)
- Gets metadata (filename,source)

Build Vector Stores

- Chunking (break large docs)
- Embedding (OpenAI)
- Chroma: Persistent on disk
- FAISS: faster memory, fast session level

Retrieve Context

- Searches 4 vector Stores
- SCU KB and User Upload: each source has 2 stores (Chroma and FAISS)
- De-duplicates results, context to LLM
- Retrieval Priority: Primary KB (Chroma), Backup KB(FAISS),User Uploaded (Chroma and FAISS)

CREW AI(Agents & Tools)

Get_Crew_Agent_Tool (personality)

- Context about SCU
- Tone and style
- Drahmann Center fallback (Safety+responsibility)
- Safety guardrails (Hallucination and wrong advice)

Is_scu_related_question (function)

- Filter
- SCU Question (KB first)
- General (Fall back to websearch)

Run_Agent_Answer (generation)

- Task: Ties everything together
  1) Carefully read the user's questions
  2) Determine if we need to use web search:
     a) For SCU questions: use KB first. If insufficient, search site: scu.edu
     b) For general questions: use web search
  3) Answer accurately and cite all sources
- Search guidance
- Drahmann Center fallback

Features

Voice Input

Talk to chatbot instead of typing

Chat History

Preserves users previous questions

KB Indexing

Enables faculty to add or remove documents and re-vectorize documents immediately, ensuring the most up to date information

End User Uploads

End users can upload transcripts, 4 year plans and other documents to reference in queries

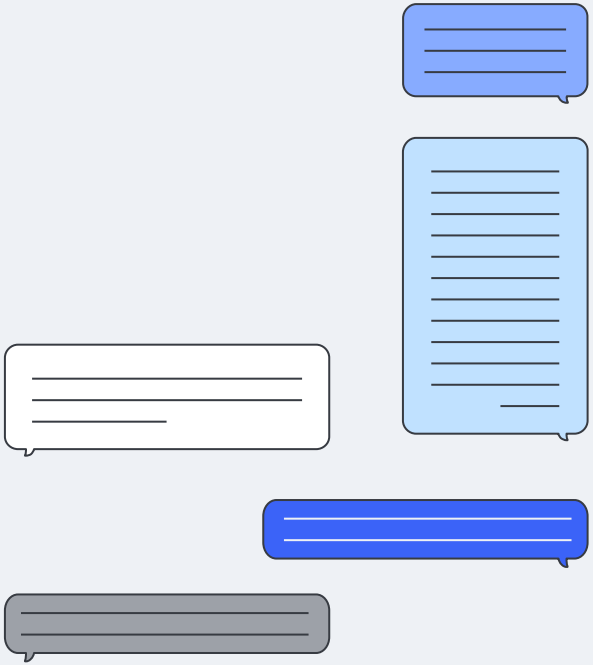
Dark/ Light Mode

User centric customization, allowing users to customize if they want dark or light mode

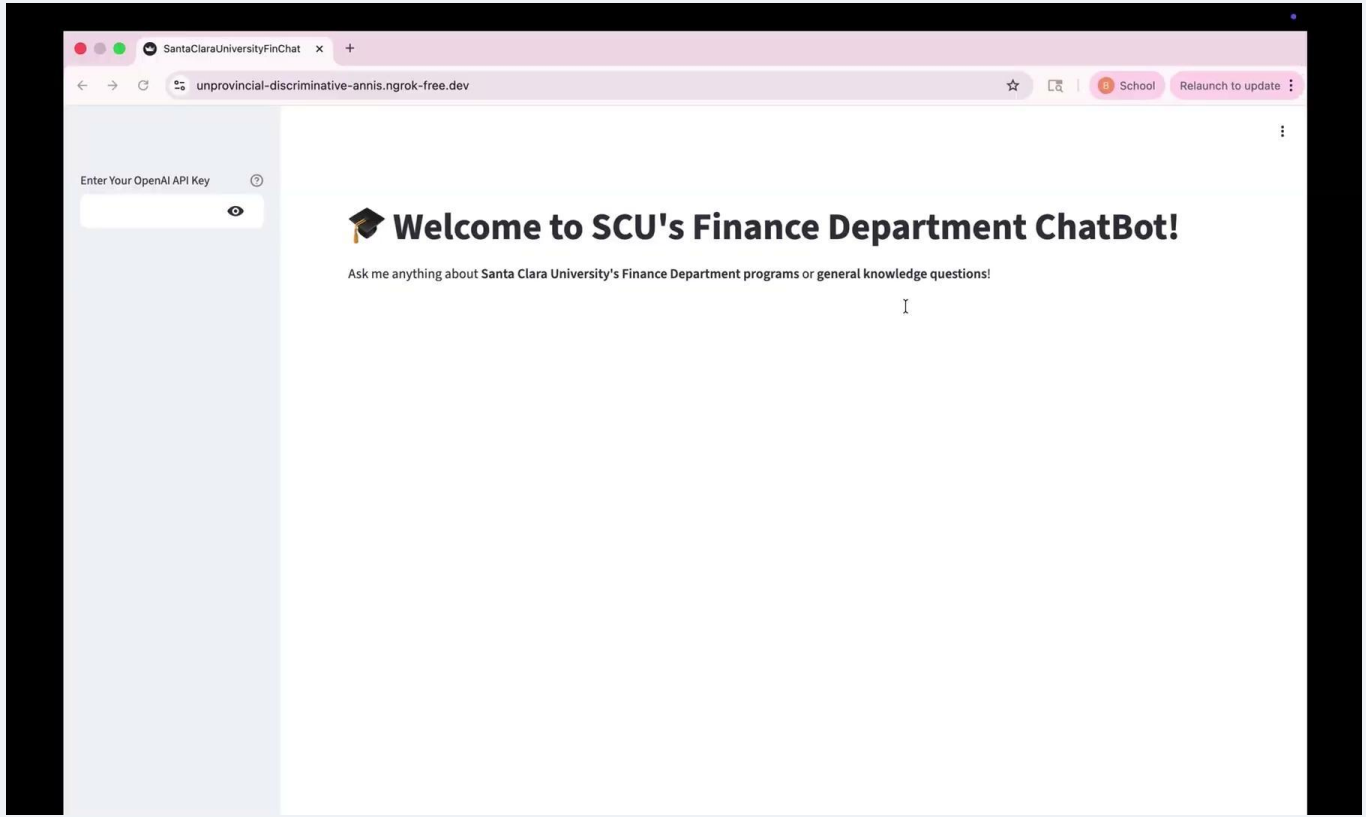
Web Search

It gives the system the ability to answer questions outside local dataset by pulling verified information from SCU websites or the broader internet when needed.

UI Overview

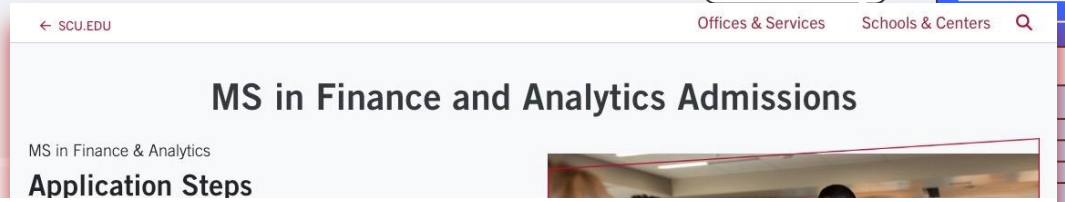


Implementation : Demo Video



DEMO VIDEO

Implementation:



what are the prerequisites to enroll in SCU finance department

To enroll in the Finance program at Santa Clara University's Leavey School of Business, the following prerequisites are generally required for undergraduate students:

1. **General Course Requirements**:
   - Completion of foundational courses in Mathematics. Students can choose from:
     - MATH 12 (Calculus for Business and Social Science)
     - MATH 31 (Calculus for Business)
2. **Core Finance Courses**:
   - Some core finance courses may have prerequisites, such as successful completion of FNCE 121 (Principles of Finance) before advancing to more specialized finance courses.
3. **Overall Academic Standing**:
   - Students typically need to be juniors or seniors in their academic standing to declare a major in Finance.
4. **Grade Requirement**:
   - A minimum grade may be required in all prerequisite courses to proceed in the Finance curriculum.

For specific details, including any changes to the requirements, I recommend visiting the official SCU Leavey School of Business Finance Academics page or checking the Finance Checklist.

Ask anything - SCU Finance Department or general knowledge...

Implementation: Safety Guardrails

Shut the(¬◡¬)up

I'm here to assist with academic-related questions, particularly regarding Santa Clara University's Finance Department programs or finance-related topics. If you have any inquiries or need information on those subjects, feel free to ask! Otherwise, I won't be able to respond to inappropriate language or behavior. Thank you for understanding.

I want to pew pew someone

I'm sorry, but I can't assist with that request. If you're feeling angry or upset, I strongly encourage you to speak with someone who can help, such as a mental health professional or a trusted person in your life. My purpose is to provide academic assistance only.

Implementation: End User Doc Upload

can I add real estate minor and still graduate in 4 years

Based on the context provided, it's feasible to add a Real Estate Minor and still graduate in four years, but it is advisable to consult with the Drahmann Academic Advising Center for personalized guidance and to confirm your specific academic plan.

View Local Knowledge Base Sources

[1] Source: `/tmp/tmpwp0dlwyp.pdf`

Sample Four-Year PlanStudent Name: Student ID#: Major: Minor:
First Year
Quarter 1 - Fall Quarter 2 - Winter Quarter 3 - Spring Summer YearCumulativeCourse NameCategoryDivisionUnits Course NameCategoryDivisionUnits Course NameCategoryDivisionUnits Course NameCategoryDivisionUnits BUSN 70 BUSN CoreLower4 ENGL 2A SCU CoreLower4 ECON 2 BUSN CoreLower4ENGL 1A SCU CoreLower4 Cultures & Ideas 1SCU CoreLower4 Cultures & Ideas 2SCU CoreLower4MATH 9 or 30BUSN CoreLower4 ECON 1 BUSN CoreLower4 Diversity SCU CoreLower4RTC I SCU CoreLower4 OMIS 15 BUSN CoreLower2 OMIS 34 BUSN CoreLower4MGMT 71 BUSN CoreLower2
Lower Division: 16 Lower Division: 16 Lower Division: 16 Lower Division: 0 48 48Upper Division: 0 Upper Division: 0 Upper Division: 0 Upper Division: 0 0 0Total: 16 Total: 16 Total: 16 Total: 0 4...

Future Enhancements

- Move from ngrok to AWS for long-term hosting
- Enable students to read, upload and answers questions based on images (jpg, jpeg files)
- More mobile friendly
- Provide three automatic starter questions to help users begin the conversation effortlessly, without needing to type a prompt
- Integration with SCU workday interface to pull even more personalized customization
- Google calendar integration for important deadlines such as add/drop etc.
- Multi language support for international students

Thank You!